

Weekly Report

SVM (Support Vector Machine)

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1 Background

Do not be afraid of its name. Its is just a algorithm to find the best boundary between two classes. I think that recent algorithms are named complexly, but the basic idea is simple. Before the main idea, let's consider three concepts:

1. Hyperplane

Based on the d dimensional space, the hyperplane that functions as a boundary is defined in $d - 1$ dimensional space.

2. Support Vector

Each point (named as vector) in the dataset that is closest to the hyperplane.

3. Margin

The distance between the hyperplane and the support vector.

Therefore, the objective of SVM is to find the hyperplane that maximizes the margin between two classes [1]

2 Equation

let's define the two classes, X, Y , and its data in Equation 1.

$$\begin{aligned} X &:= \{x_1, x_2, \dots, x_n\}, & x_i &\in \mathbb{R}^d \\ Y &:= \{y_1, y_2, \dots, y_m\}, & y_i &\in \mathbb{R}^d \end{aligned} \tag{1}$$

As a hyperplane, $w \in \mathbb{R}^d, b \in \mathbb{R}$, the basic constraints are given by the following equation (WLOG) in Equation 2.

$$\begin{aligned} w^T x + b &\geq 1, & \forall x \in X \\ w^T y + b &\leq -1, & \forall y \in Y \end{aligned} \tag{2}$$

Let's consider the margin from class X . the distance between hyperplane, $w^T x + b$, is defined as following equation in Equation 4.

$$d_{x,i} := \frac{|w^T x_i + b|}{\|w\|} \tag{3}$$

and the margin of class X is defined as the minimum distance from all points in class X in Equation 4.

$$\begin{aligned} m_X &:= \min_{i \in \{1, \dots, n\}} d_{x,i} \\ &\leq \frac{|1|}{\|w\|} = \frac{1}{\|w\|} \end{aligned} \tag{4}$$

Correspondingly, the margin of class Y is defined as the minimum distance from all points in class Y in Equation 5.

$$\begin{aligned} d_{y,i} &:= \frac{|w^T y_i + b|}{\|w\|} \\ m_Y &:= \min_{i \in \{1, \dots, m\}} d_{y,i} \\ &\leq \frac{|1|}{\|w\|} = \frac{1}{\|w\|} \end{aligned} \tag{5}$$

Therefore, the optimization equation is defined as the following equation in Equation 6.

$$\begin{aligned} \max_{w,b} \quad & m_X + m_Y = \frac{2}{\|w\|} \\ \text{s.t.} \quad & w^T x + b \geq 1, \quad \forall x \in X \\ & w^T y + b \leq -1, \quad \forall y \in Y \end{aligned} \quad (6)$$

Therefore, maximizing the margin $\frac{2}{\|w\|}$ is equivalent to minimizing $\frac{1}{2}\|w\|^2$, leading to the formulation in Equation 7.

$$\begin{aligned} \min_{w,b} \quad & \frac{1}{2}\|w\|^2 \\ \text{s.t.} \quad & w^T x + b \geq 1, \quad \forall x \in X \\ & w^T y + b \leq -1, \quad \forall y \in Y \end{aligned} \quad (7)$$

3 Optimization process (Lagrange multiplier)

I will additionally write the report relative to the Lagrange multiplier method after L-BFGS method. In this time, just merely look the format of loss function (sorry). The loss function to transform the constraints to unconstrained is defined as the following equation in Equation 8. if system violate constraints in Equation 7, these component increase the loss function. So, the loss function should be minimized.

$$\begin{aligned} L(w, b, \alpha, \beta) = & \frac{1}{2}\|w\|^2, \quad \leftarrow \text{objective} \\ & - \sum_{i=1}^n \alpha_i (w^T x_i + b - 1) \quad \leftarrow \text{inequality constraints, } \alpha_i \geq 0 \\ & + \sum_{j=1}^m \beta_j (w^T y_j + b + 1) \quad \leftarrow \text{inequality constraints, } \beta_j \geq 0 \end{aligned} \quad (8)$$

Maybe you know that to find the local minimum or local maximum, the value of gradient should be zero for all variables. Therefore, so, we should find the values that satisfy the following equation in Equation 9.

$$\begin{aligned} \frac{\partial L}{\partial w} = 0 = w - \sum_{i=1}^n \alpha_i x_i + \sum_{j=1}^m \beta_j y_j & \Rightarrow w = \sum_{i=1}^n \alpha_i x_i - \sum_{j=1}^m \beta_j y_j \\ \frac{\partial L}{\partial b} = 0 = - \sum_{i=1}^n \alpha_i + \sum_{j=1}^m \beta_j & \Rightarrow \sum_{i=1}^n \alpha_i = \sum_{j=1}^m \beta_j \end{aligned} \quad (9)$$

Ok, to apply the hint from Equation 9, the loss function is transformed to the following equation in Equation 10.

$$\begin{aligned}
L(w, b, \alpha, \beta) &= \frac{1}{2} \|w\|^2 - \sum_{i=1}^n \alpha_i (w^T x_i + b - 1) + \sum_{j=1}^m \beta_j (w^T y_j + b + 1) \\
&= \frac{1}{2} \|w\|^2 - \left(\sum_{i=1}^n \alpha_i w^T x_i \right) - \left(\sum_{i=1}^n \alpha_i b \right) + \left(\sum_{i=1}^n \alpha_i \right) + \left(\sum_{j=1}^m \beta_j w^T y_j \right) + \left(\sum_{j=1}^m \beta_j b \right) + \left(\sum_{j=1}^m \beta_j \right) \\
&= \frac{1}{2} \|w\|^2 - w^T \left(\sum_{i=1}^n \alpha_i x_i - \sum_{j=1}^m \beta_j y_j \right) - b \left(\sum_{i=1}^n \alpha_i - \sum_{j=1}^m \beta_j \right) + \left(\sum_{i=1}^n \alpha_i + \sum_{j=1}^m \beta_j \right) \\
&= \frac{1}{2} \|w\|^2 - w^T w + \left(\sum_{i=1}^n \alpha_i + \sum_{j=1}^m \beta_j \right) \\
&= \left(\sum_{i=1}^n \alpha_i + \sum_{j=1}^m \beta_j \right) - \frac{1}{2} \|w\|^2
\end{aligned} \tag{10}$$

In this equation, the sloop term can be expressed as the following equation in Equation 11.

$$\begin{aligned}
\|w\|^2 &= w^T w \\
&= \left(\sum_{i=1}^n \alpha_i x_i - \sum_{j=1}^m \beta_j y_j \right)^T \left(\sum_{i=1}^n \alpha_i x_i - \sum_{j=1}^m \beta_j y_j \right) \\
&= \sum_{i=1}^n \sum_{k=1}^n \alpha_i \alpha_k x_i^T x_k - 2 \sum_{i=1}^n \sum_{j=1}^m \alpha_i \beta_j x_i^T y_j + \sum_{j=1}^m \sum_{l=1}^m \beta_j \beta_l y_j^T y_l
\end{aligned} \tag{11}$$

Therefore, the loss function can be expressed without w and b as the following equation in Equation 12.

$$\begin{aligned}
L(w, b, \alpha, \beta) &= L(\alpha, \beta) \\
&= \left(\sum_{i=1}^n \alpha_i + \sum_{j=1}^m \beta_j \right) - \frac{1}{2} \left(\sum_{i=1}^n \sum_{k=1}^n \alpha_i \alpha_k x_i^T x_k - 2 \sum_{i=1}^n \sum_{j=1}^m \alpha_i \beta_j x_i^T y_j + \sum_{j=1}^m \sum_{l=1}^m \beta_j \beta_l y_j^T y_l \right)
\end{aligned} \tag{12}$$

To briefly describe the loss function, the objective of the first term is to minimize the sum of α_i and β_j . The second term is to maximize the distance between two classes. Then, how to find the optimal α_i and β_j ? Since α_i and β_j can not be defined by x_i and y_j , we can apply SMO (Sequential Minimal optimization) algorithm as example.

4 SMO (Sequential Minimal Optimization)

The basic idea of SMO is that **We can solve if the variables are two with fixed other variables**. Therefore, the SMO algorithm is to find the optimal two variables and fix the other variables. Then, repeat this process until convergence.

Of course, various optimization methods except SMO can be applied.

Algorithm 1 Sequential Minimal Optimization (SMO) for Dual SVM

Require: Training data (X, Y) , tolerance ϵ , max iterations T .

Ensure: Optimal dual variables α^*, β^* , weight vector w^* , bias b^* .

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1: Set  $\alpha_i \leftarrow 0$  ( $\forall i = 1, \dots, n$ ) ▷ Initialize Class X dual variables
2: Set  $\beta_j \leftarrow 0$  ( $\forall j = 1, \dots, m$ ) ▷ Initialize Class Y dual variables
3: Set  $t \leftarrow 0$ 
4: while  $t < T$  and not converged do
5:   Select arbitrary  $\alpha_i$  and  $\beta_j$  ▷ Select variable pair
6:   Set  $C \leftarrow \sum_{k \neq i} \alpha_k - \sum_{l \neq j} \beta_l$  ▷ Compute constant from equality constraint
7:   Express  $\beta_j(\alpha_i) \leftarrow \alpha_i + C$  ▷ Unify into single independent variable  $\alpha_i$ 
8:   Reformulate  $L(\alpha, \beta)$  as 1D objective  $L_1(\alpha_i)$  ▷ Redefine objective function
9:   Find  $\alpha_i^{\text{new}}$ , s.t.  $\alpha_i \geq 0$  ▷ Compute unconstrained optimum
10:  Set  $\beta_j^{\text{new}} \leftarrow \alpha_i^{\text{new}} + C$  ▷ Update paired variable  $\beta_j$ 
11:  if  $L(\alpha_i^{\text{new}}, \beta_j^{\text{new}}) - L(\alpha_i, \beta_j) < \epsilon$  then
12:    break ▷ Check convergence condition
13:  end if
14:  Set  $t \leftarrow t + 1$ 
15: end while
16: Set  $w^* \leftarrow \sum_{i=1}^n \alpha_i^* x_i - \sum_{j=1}^m \beta_j^* y_j$  ▷ Reconstruct optimal weight vector
17: Set  $b^* \leftarrow 1 - (w^*)^T x_{sv}$  ▷ Compute bias using Support Vector  $x_{sv}$ 
```

5 Expansion

The basic classification is binary classification. To expand multi-class, variable extensions from SVM are used.

References

- [1] J. Cervantes, F. Garcia-Lamont, L. Rodríguez-Mazahua, A. Lopez, A comprehensive survey on support vector machine classification: Applications, challenges and trends, *Neurocomputing* 408 (2020) 189–215.